# Moe Mirzaei (Mohammadreza Mirzaei)

## SUMMARY

An engineer with a passion for innovation. I bring a strong commitment to hard work, creative problem-solving, and sharp analytical skills. I aim to merge my technical knowledge with my passion to make a meaningful impact.

## CONTACT

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## EDUCATION

**Doctor of Philosophy, Engineering and Computing Systems** 2026-Present
Pennsylvania State University, Harrisburg

**Master's degree, Civil/Transportation Engineering (GPA: 4/4)** 2023-2025
University of Florida, Gainesville

**Bachelor's degree, Civil Engineering (GPA: 3.4/4)** 2017-2021
Sharif University of Technology, Tehran

## INTERESTS

- Intelligent Transportation Systems (ITS)
- Traffic Operations and Safety
- Artificial Intelligence and Data Science applications
- Computer Vision and Image Processing

## SKILLS

**Programming languages:**
Proficient in **Python, R** Familiar with Julia, MATLAB

**Code editors:**
Visual Studio Code, RStudio, Jupyter Notebook, PyCharm

**Database**: MongoDB, SQL

**OS**: Windows, Linux (Ubuntu)

**Software:**
Microsoft Office (Proficient in Word, Excel, and PowerPoint)
Docker, SPSS, Mathcad, GAMS

**Simulators:** HCS, TSIS CORSIM, TRANSYT-7F, SwashSim, VISSIM, VISUM, ArcGIS Pro

**Soft skills:** Critical thinking, Problem-solving, Teamwork, Project management

## LEADERSHIP

**Director of Communication of Penn State ITE** Oct 2026 – Present
*Institute of Transportation Engineers student chapter, Pennsylvania State University*

**Vice President of Gator ITE** Apr 2023 – Apr 2024
*Institute of Transportation Engineers student chapter, University of Florida*

**President of Civil Engineering Club (Scientific Association)** Jun 2019 - Aug 2020
*Sharif University of Technology*

## LANGUAGES

English, Persian, and Azerbaijani

## ACADEMIC EXPERIENCES

**Graduate Research Assistant** Aug 2026 – Present

*Dr. Agnimitra Sengupta, Pennsylvania State University Harrisburg*

- Research on traffic safety and operation, and AI applications in transportation engineering

**Graduate Research Assistant (I-STREET Lab)** Aug 2023 – May 2025

*Dr. Lily Elefteriadou & Dr. Pruthvi Manjunatha, University of Florida Transportation Institute*

**Title**: ***Evaluation of Gainesville Pedestrian-Bicyclists Connected Vehicle Pilot***

- Evaluated the passive pedestrian detection system, CV communication reliability, and user perceptions of the smartphone application by conducting focus groups and participant experiments.
- Worked with Roadside Units, On-board units, TravelSafely/Glance application, and Bosch cameras.
- Collaborated with several vendors (Yunex, Applied Information, Bosch, etc.) and FDOT.

**Research (Remote)** May 2022 – Apr 2023

*Dr. Kara Kockelman, University of Texas at Austin*

**Title:** *Inferring vehicle speeds, makes and models from video camera footage: Ready opportunities for transport data analysis.*

**Research Assistant** Aug 2020 – Sep 2021

*Dr. Amir Samimi, Sharif University of Technology*

➢ Researched Urban Freight Transportation types and algorithms for optimizing **Vehicle Routing Problems (VRP) and Pickup and Delivery Problems (PDP)**. Designed and developed a dashboard for solving several types of VRP with constraints using two algorithms (*Genetic Algorithm* and *Simulated Annealing*) in ***Python* and *R*** programming languages.

**Teaching Assistant**

- Transportation Engineering Dr. Pruthvi Manjunatha, University of Florida — Oct 2024
- Python Programming Dr. Elham Seifossadat, Sharif University of Technology — Feb 2020 – Oct 2021
- Engineering Probability and Statistics Dr. Hatef Abdoos, Sharif University of Technology — Sep 2019 – Dec 2019

## WORK EXPERIENCES

**ITS Traffic Engineer** Jun 2025 – Mar 2026

*AtkinsRealis, Florida Department of Transportation District Two (FDOT D2)*

In this role, I support the FDOT D2 on a range of ITS initiatives, including Connected Vehicle deployments, innovative Truck Parking Availability System (iTPAS), Rail Crossing Detection, Wrong Way Vehicle Detection, Video Analytics, and other emerging technology evaluations.

**Transportation Data Analyst (Internship)** Jul 2020 - Aug 2020

*Roads and Transportation Information Management Center*

1) Researched the information provided by traffic cameras in different countries.
2) Troubleshooted the location and algorithm of travel time cameras.
3) Detected and troubleshooted traffic counters in three-ways.

## CERTIFICATES

**IMSA Traffic Signal Technician I** Credential ID AA_150495 — Mar 2026 - Mar 2029

*IMSA-International Municipal Signal Association* — *Credential ID AA_150495*

**Advanced Temporary Traffic Control (TTC) Maintenance of Traffic (MOT)** — Oct 2025- Oct 2029

*Florida Department of Transportation* — *Credential ID 649038*

**NEMA Troubleshooting Program** — Oct 2025

*East Coast Training Alliance*